

PHYSIOLOGY OF RESPIRATION

Respiration:

- Respiration is the movement of air in and out of the lungs.
- The main purpose of respiration is to supply O₂ to the cells for their metabolic needs and to remove CO₂ from cells.

Respiration can be divided into three operational stages:

A. Internal Respiration or Cellular Respiration:

Refers to the metabolic processes carried out within the mitochondria, which use O₂ and produce CO₂, while deriving energy from nutrient molecules.

B. Intermediate Respiration:

-Includes the vascular transport (hemoglobin transport) of O₂ and CO₂ between the lung and tissues cells of the body.

C. External Respiration:

Refers to the sequence of events involved in the exchange of O₂ and CO₂ between the external environment and the cells of the body

A. **Ventilation** – movement of air into and out of the lungs.

B. O₂ and CO₂ are exchanged between air in alveoli and blood within the pulmonary capillaries by means of diffusion.

C. O₂ and CO₂ are exchanged between tissues and blood by process of diffusion across systemic (tissue) capillaries.

Non respiratory Functions of the Respiratory System:

1. Route for water loss and heat elimination.
2. Enhances venous return.
3. Helps maintain normal acid-base balance.
4. Enables speech, singing, and other vocalizations.
5. Defends against inhaled foreign matter.
6. Removes, modifies, activates, or inactivates various materials passing through the pulmonary circulation.
7. Nose serves as the organ of smell.

The Respiratory System Consists of:

- A. The lungs.
- B. The respiratory airways.
- C. The respiratory muscles.

Functions of the Respiratory Airways:

1. Conduct air to and from the lungs gas exchange surface (alveoli).
2. Aid in the warming and humidification of the inhaled air.

Large particles are usually filtered out in the nose, whereas inhaled particles of 2 to 10 μm may settle or adhere to the mucous that lines the bronchi.

Cycle of respiration or pulmonary ventilation

This occurs **12 to 15 times** per minute and consists of three phases:

- A. *Inspiration*
- B. *Expiration*
- C. *Pause.*

MECHANISM OF INSPIRATION

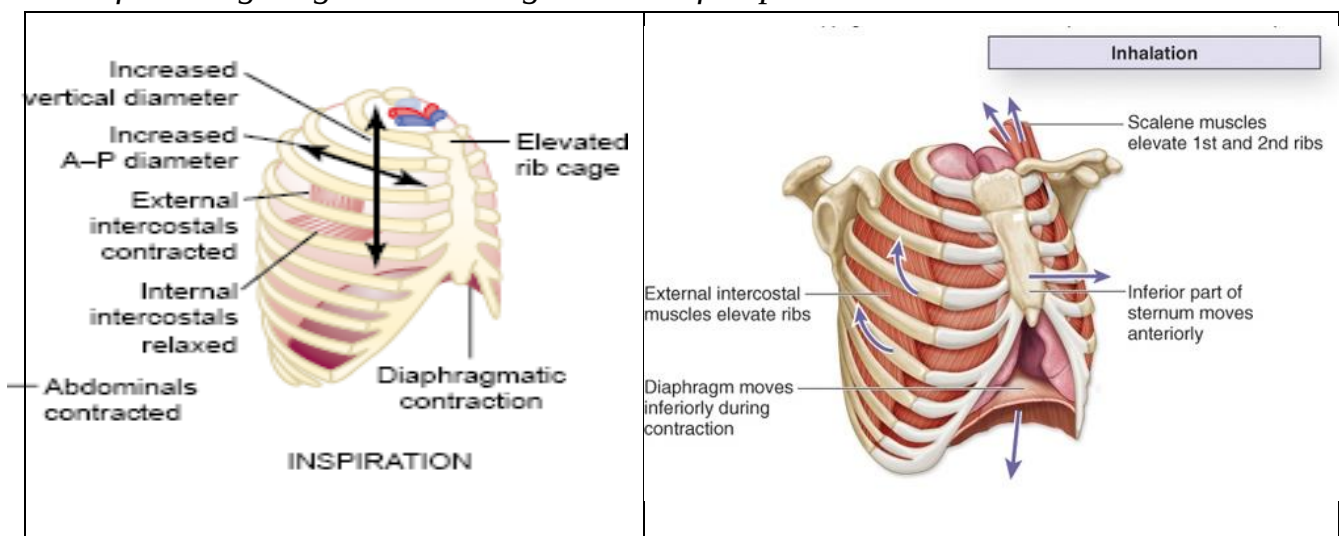
★ During Normal Quiet inspiration:

- is an active process; caused by contraction of the *diaphragm & external intercostal muscles*
- **The Diaphragm** is the main muscle of inspiration (75% of inspiratory act) → increase in the vertical diameter of thoracic cavity
- **External intercostal muscles** contraction → increase in the lateral dimensions of thoracic cavity

★ During Forced Inspiration:

- Forced inspiration is aided by **accessory muscles** of inspiration which include:
- **sternomastoid** (elevation of the sternum)
- **scalene muscles** (elevation of upper ribs)

-The following diagrams showing muscles of inspiration and their role.



MECHANISM OF EXPIRATION

★ During Normal Quiet expiration:

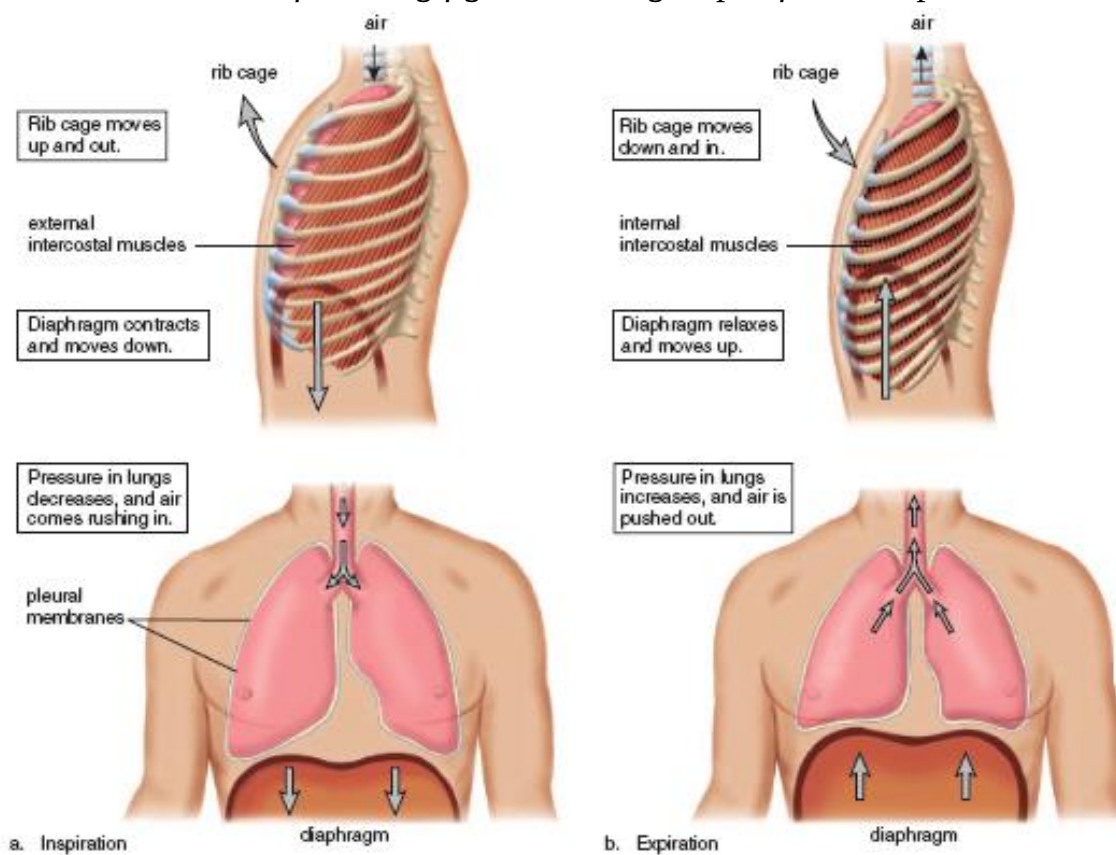
- is a passive process occurring as a result of:

- relaxation of muscles of inspiration
- Elastic recoil of the lungs
- Descent of thoracic cage by gravity
- Leading to decrease in volume of the lungs → *increase in intrapulmonary pressure* to be greater than atmospheric pressure (+1mm Hg) → pressure gradient → rush of air out of the lungs

★ **During Forced expiration:**

- is an *active process* caused by contraction of muscles of forced expiration;
- abdominal muscles
- internal intercostal muscles
- Contraction of these muscles leads to greater decrease in lung volume and hence greater pressure gradient between intrapulmonary & atmospheric_P leading to more air moving out of the lungs

The following figure showing steps of both inspiration and expiration



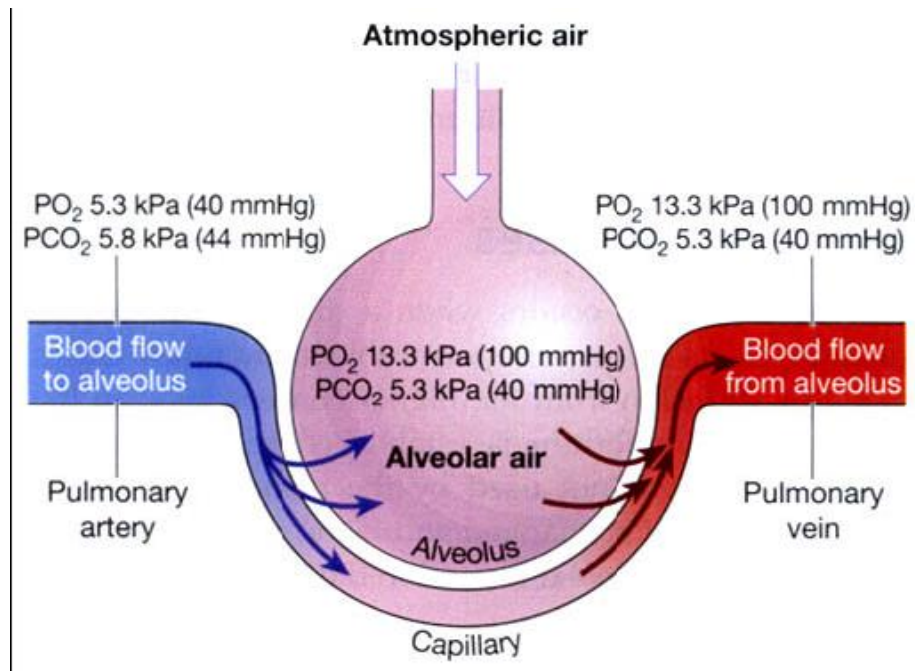
External Respiration

Definition: is exchange of gases *by diffusion* between the *alveoli and the blood*.

- Venous blood contains high levels of CO₂ and low levels of O₂.
- CO₂ diffuses from venous blood into the alveoli.
- O₂ diffuses from the alveoli into the blood.

- When blood leaves the alveolar capillaries, the O_2 and CO_2 concentrations are in equilibrium with those of alveolar air

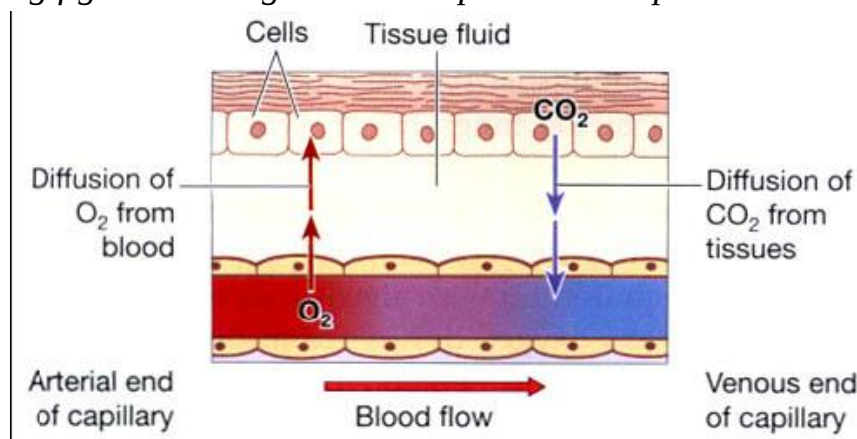
The following figure showing how is this process take place.



Internal Respiration

Definition: is exchange of gases by *diffusion* between *blood and the body cells*.

- Gaseous exchange does not occur across the walls of the arteries because their walls are too thick.
- Blood arriving at the tissues has been cleansed of its CO_2 and saturated with O_2 and therefore has a *higher* PO_2 and a *lower* PCO_2 than the tissues.
- O_2 diffuses from the bloodstream through the capillary wall into the tissues.
- CO_2 diffuses from the cells into the extracellular fluid then into the bloodstream towards the venous end of the capillary.
- The following figure showing how is this process take place.



Transport of Gases in the Bloodstream

Transport of blood oxygen and carbon dioxide is essential for internal respiration to occur.

I: Oxygen is carried by blood in 2 forms:

1: Physically: dissolved in water of the plasma

0.3ml / 100 ml blood. (1.5% of blood O₂).

2: Chemical form with Hb: As oxyhaemoglobin (98.5% of blood O₂)

- Oxyhaemoglobin dissociates releasing oxygen.

Amount of O₂ carriage:

= 20 ml / 100 ml blood in arterial blood.

= 14 ml / 100 ml blood in venous blood.

- Each molecule of Hb combines with 4 molecules of O₂, by oxygenation and not oxidation, on the ferrous iron not ferric iron. (*each Hb molecule contains 4 atoms of ferrous iron*)

- Each gram of Hb can combine with 1.34 ml O₂.

100ml of blood contain, normally, 15gm of Hb.

O₂ capacity = 1.34 X Hb content = 1.34 X 15 = 20 ml O₂ / dl

It is the maximum volume of oxygen carried by 100ml blood.

O₂ content = 1.34 X Hb content at different P O₂ (at 100 mmHg=19ml / dl) , (while at 40 mmHg=14ml / dl).

It is the volume of oxygen carried by 100ml blood.

Percentage saturation (% O₂ Saturation) = O₂ content / O₂ capacity X100.

* % saturation of O₂ in arterial blood = 19 / 20 X100= **95%**

* % saturation of O₂ in venous blood = 14 / 20 X100= **70%**

Factors that increase dissociation include:

A. raised carbon dioxide content of tissue fluid

B. raised temperature

C. Increased 2,3-diphosphoglycerate (2,3-DPG is a substance present in red blood cells).

- In active tissues there is increased production of carbon dioxide and heat which leads to increased release of oxygen. In this way oxygen is available to tissues in greatest need. When oxygen leaves the erythrocyte, the deoxygenated hemoglobin turns purplish in color.

II.CO₂ Carriage

CO₂ in the arterial blood = 48ml / dl in 2 forms:

1: Physically dissolved in the plasma = 3 ml / dl blood. (7% of blood CO₂)

2: Chemical combination = 45 ml / dl blood as:

A) Bicarbonate = 42 ml / dl, in RBCs (K HCO_3), and plasma (Na HCO_3). (70% of blood CO_2)

B) carbamino compounds = 3 ml / dl in RBCs (globins), and plasma (plasma proteins). (23% of blood CO_2).

Lung volumes and capacities:

Determination of lung volumes is used to:

1-assess the efficiency of the respiratory system.

2-diagnose respiratory diseases.

- Most of these volumes can be measured using a simple *spirometer*

Pulmonary volumes:

• **Tidal volume (TV):**

It is the volume of air inspired or expired by each breath during normal quiet breathing. It is about 500 ml.

• **Inspiratory reserve volume (IRV)**

It is maximal volume of air which can be inspired after a normal inspiration. It is about 3000 ml.

• **Expiratory reserve volume (ERV)**

It is the maximal volume of air which can be expired after a normal expiration. It is about 1100 ml.

• **Residual volume (RV)**

It is the volume of air remaining in the lungs after maximal expiration. It is about 1200 ml.

• **Pulmonary Capacities**

A capacity is two volumes or more added together.

1- **Inspiratory capacity (IC)**

It is the maximal volume of air that can be inspired from the resting expiratory volume. It is about 3500 ml

$$IC = TV + IRV$$

$$500 + 3000 = 3500 \text{ ml.}$$

2- **Functional residual capacity (FRC)**

It is the volume of air which remains in the lung at the resting expiratory level (after normal expiration). It is also called relaxation volume. It is about 2300 ml.

$$FRC = RV + ERV$$

$$= 1200 + 1100 = 2300 \text{ ml}$$

3- Vital capacity (VC)

It is the maximum volume of air that can be expelled from lung by a maximal expiration after a maximal inspiration.

It is about 4600 ml

$$VC = IRV + TV + ERV$$

$$3000 + 500 + 1100 = 4600$$

It is a good index for pulmonary efficiency.

4- Total lung capacity (TLC)

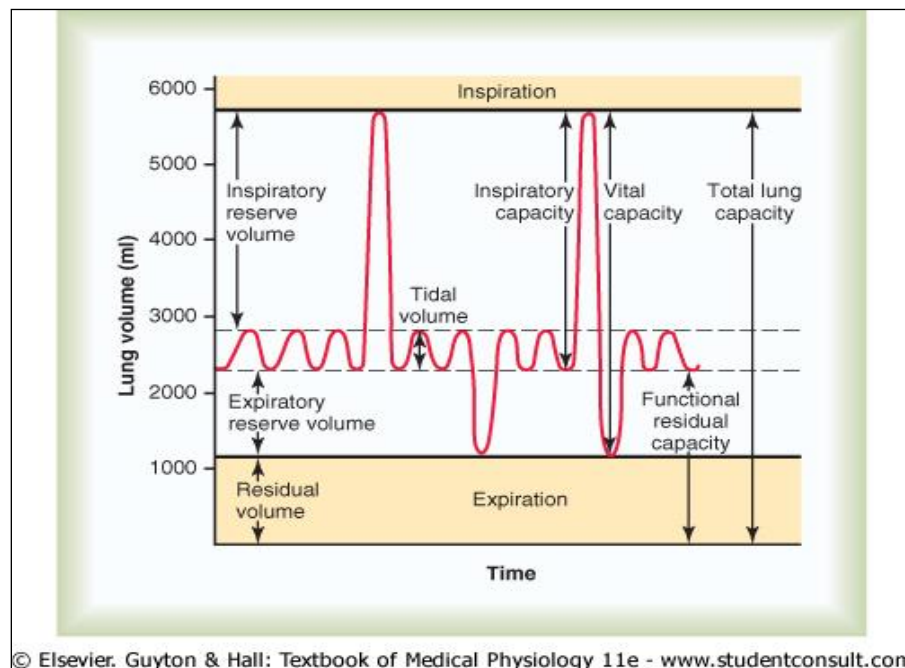
It is the volume of air contained in the lung at the end of maximal inspiration.

It is about 5800 ml.

$$TLC = IRV + TV + ERV + RV$$

$$= 3000 + 500 + 1100 + 1200 = 5800 \text{ ml.}$$

The following figure showing the previous Lung volumes and capacities



Dead space

It is the volume of air which does not undergo gas exchange with pulmonary capillaries.

Types of dead space:

- A. Anatomical dead space.
- B. Alveolar dead space .
- C. Physiological dead space.

Measurement of dead space

By using Bohr's equation (physiological DS):

$$DS = TV \times \frac{PCO_2 \text{ in alveolar air} - PCO_2 \text{ in expired air}}{PCO_2 \text{ in alveolar air}}$$

PCO₂ in arterial blood 40mmHg

PCO₂ in expired air 28mmHg

T.V 500ml

$$= 500 \times \frac{40 - 28}{40}$$

Control of Respiration

- Sensory information (PCO₂, lung stretch, irritants, muscle spindles, tendons, and joints) is coordinated in the brain stem.
- The output of the brain stem controls the respiratory muscles and the breathing cycle.

A. Central control of breathing (brain stem and cerebral cortex)

1. *Medullary respiratory center*

- *is located in the reticular formation,*

a. *Dorsal respiratory group*

- is primarily responsible for inspiration and generates the basic rhythm for breathing.
- Input to the dorsal respiratory group comes from the vagus and glossopharyngeal nerves. The vagus nerve relays information from peripheral chemoreceptors and mechanoreceptors in the lung. The glossopharyngeal nerve relays information from peripheral chemoreceptors.
- Output from the dorsal respiratory group travels, via the phrenic nerve, to the diaphragm.

b. *Ventral respiratory group*

- is primarily responsible for forced expiration and inspiration.
- is not active during normal, quiet breathing, when expiration is passive.
- is activated, for example, during exercise, when expiration becomes an active process.

2. *Apneustic center*

- is located in the **lower pons**.
- **stimulates inspiration**, producing a deep and prolonged inspiratory gasp (apneusis).

3. *Pneumotaxic center*

- is located in the **upper pons**.
- **inhibits inspiration** and, therefore, regulates inspiratory volume and respi-

ratory rate.

4. Cerebral cortex

- Breathing can be under voluntary control; therefore, a person can voluntarily hyperventilate or hypoventilate.
- Hypoventilation (breath-holding) is limited by the resulting increase in PCO_2 and decrease in PO_2 . A previous period of hyperventilation extends the period of breath-holding.
- The following figure showing organization of the respiratory center

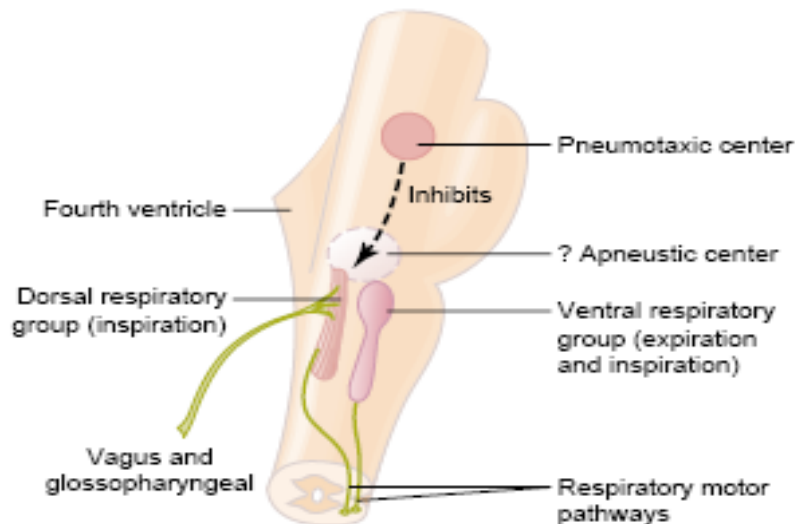


Figure 41-1

Organization of the respiratory center.

B. Chemoreceptors for CO_2 , H^+ , and O_2

1. Central chemoreceptors in the medulla

- are sensitive to the **pH** of the cerebrospinal fluid (CSF). Decreases in the pH of the CSF produce increases in breathing rate (hyperventilation).
- H^+ does not cross the blood-brain barrier as well as CO_2 does.
- a. CO_2 diffuses from arterial blood into the CSF because CO_2 is lipid-soluble and readily crosses the blood-brain barrier.
- b. In the CSF, CO_2 combines with H_2O to produce H^+ and HCO_3^- . The resulting **H^+ acts directly on the central chemoreceptors.**
- c. Thus, increases in PCO_2 and $[\text{H}^+]$ stimulate breathing, and decreases in PCO_2 and $[\text{H}^+]$ inhibit breathing.
- d. The resulting hyperventilation or hypoventilation then returns the arterial PCO_2 toward normal.

2. Peripheral chemoreceptors in the carotid and aortic bodies

- The carotid bodies are located at the bifurcation of the common carotid

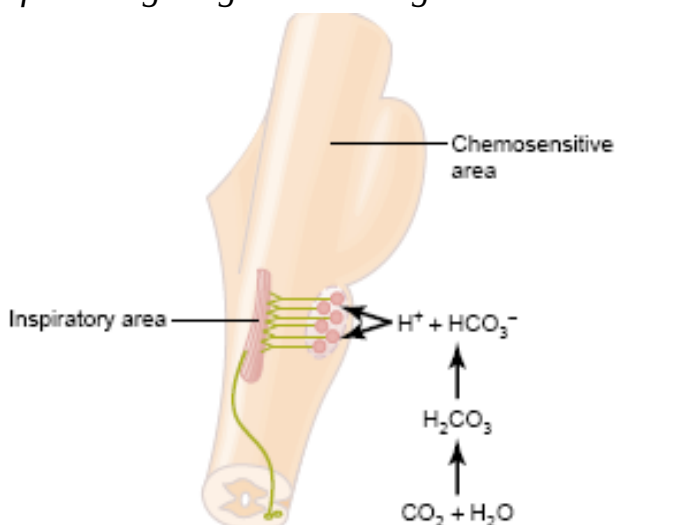
arteries.

- The aortic bodies are located above and below the aortic arch,
- a. **Decreases in arterial PO_2**
- stimulate the peripheral chemoreceptors and **increase breathing rate.**
 PO_2 must decrease to low levels ($< 60 \text{ mm Hg}$) before breathing is stimulated.
 When PO_2 is less than 60 mm Hg, breathing rate is exquisitely sensitive to PO_2 .
- b. **Increases in arterial PCO_2**
- stimulate peripheral chemoreceptors and **increase breathing rate.**
- potentiate the stimulation of breathing caused by hypoxemia.
- The response of the peripheral chemoreceptors to CO_2 is less important than the response of the central chemoreceptors to CO_2 (or H^+).
- c. **Increases in arterial $[H^+]$**
- stimulate the carotid body peripheral chemoreceptors directly, independent of changes in PCO_2 .
- In metabolic acidosis, breathing rate is increased (hyperventilation) because arterial $[H^+]$ is increased and pH is decreased.

TABLE 4-7 **Comparison of Central and Peripheral Chemoreceptors**

Type of Chemoreceptor	Location	Stimuli that Increase Breathing Rate
Central	Medulla	$\downarrow$ pH $\uparrow$ (PCO_2)
Peripheral	Carotid and aortic bodies	$\downarrow$ PO_2 (if $< 60 \text{ mm Hg}$) $\uparrow$ PCO_2 $\downarrow$ pH

The following diagram showing chemosensitive area



C. Other types of receptors for control of breathing

1. *Lung stretch receptors*

- are located in the smooth muscle of the airways.
- When these receptors are stimulated by distention of the lungs, they produce a reflex decrease in breathing frequency (**Hering-Breuer reflex**).

2. *Irritant receptors*

- are located between the airway epithelial cells.
- are stimulated by noxious substances (e.g., dust, pollen).

3. *J (juxtacapillary) receptors*

- are located in the alveolar walls, close to the capillaries.
- Engorgement of the pulmonary capillaries, such as may occur with **left heart failure**, stimulates the **J** receptors, which then cause rapid, shallow breathing.

4. *Joint and muscle receptors*

- are activated during movement of the limbs.
- are involved in the early stimulation of breathing during exercise.

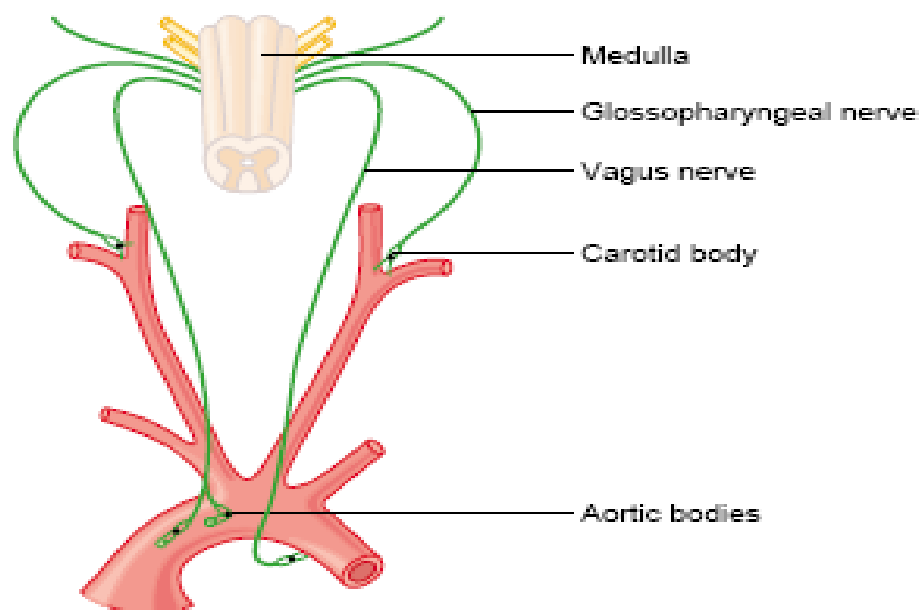


Figure 41-4

Respiratory control by peripheral chemoreceptors in the carotid and aortic bodies.

Other factors that influence respiration

-Breathing may be modified by the higher centers in the brain by:

- *speech, singing*
- *emotional displays, e.g. crying, laughing, fear*
- *drugs, e.g. sedatives, alcohol*
- *sleep.*

Temperature influences breathing.

- In fever respiration is increased due to increased metabolic rate
- In hypothermia it is depressed, as is metabolism.
- Temporary changes in respiration occur in swallowing, sneezing and coughing.